International Scientific Physics Olympiad

T1. Polycrystal diffusion

Answer sheet

Part A. Temperature dependence of the diffusion coefficient (2.6 points)

A1 **0.2**

$v =$

$D(T) \propto$

A2 **0.4**

$\Delta r_{\text{sq}} =$

A3 **2.0**

T	t	N	Δr_{sq}	D

$E_g =$

Accuracy:

Part B. Effective diffusion coefficient (2.4 points)

B1 **0.4**

$\alpha =$

B2 **2.0**

t	D_{gb}/D	δ	α

$\alpha =$

Part C. Type B kinetics (5 points)

C1 **0.4**

C2 **4.0**

t	D_{gb}/D	δ	$\ln \bar{c}$

$\xi =$ $\eta =$

$\zeta =$

C3 **0.6**

$C =$

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T2. Muon anomalous magnetic moment

Answer sheet

Part A. Dynamics in magnetic field (4.5 points)

A1 (Formula) $\Delta v =$ (Value) $\Delta v =$ **1.0**

A2 (Formula) $E_k =$ (Value) $E_k =$ **0.5**

A3 (Formula) $R =$ (Value) $R =$

(Formula) $T =$ (Value) $T =$ **1.0**

A4 (Formula) $N =$ (Value) $N =$ **0.8**

A5 $d\vec{s}/dt =$ **0.7**

A6 $\omega_z =$ **0.5**

Part B. Thomas precession (3.5 points)

B1 $v_{Cx} =$ $v_{Cy} =$ **0.5**

B2 $v''_{x''} =$ $v''_{y''} =$ **0.7**

B3 **0.6**

B4 $\omega =$ **1.0**

B5 $\omega_{T,z} =$ **0.7**

Part C. Magnetic moment precession (2 points)

C1 **0.7**

$\omega_s =$

C2 **0.5**

$\omega_a =$

C3 **0.8**

(Formula) $a_\mu =$ (Value) $a_\mu =$

T3. Frequency measurements with ring resonators

Answer sheet

Part A. Field profile in fiber optics (2.5 points)

A1 $C_1 =$ $C_2 =$ **0.3**

A2 $k_{\text{in}} =$ $k_{\text{out}} =$ **0.5**

A3 **0.3**

A4 $B =$ **0.3**

A5 **0.3**

A6 **0.3**

A7 Equation: **0.3**

A8 Equation: **0.2**

Part B. Fiber resonator transmittance (3 points)

B1 Equation: **0.2**

B2 $B_{\text{in}}(t) =$ **0.3**

B3 $B_{\text{in}}(t) =$ **0.3**

B4 $B_{\text{in}}(t) =$ **0.5**

B5 $N_2 =$ **0.5**

B6 Plot: **0.6**

B7 $Q =$ **0.6**

Part C. Calibration of the Mach-Zehnder Interferometer (2 points)

C1 Plot: **0.5**

C2 Plot: **1.0**

C3 $\Delta\omega_{\mathrm{MZI}}/\omega_{\mathrm{MZI}} =$ **0.5**

Part D. Cubic nonlinearity and solitons (2.5 points)

D1 $v_g =$ **0.1**

D2 **0.6**

D3 $D_1 =$ $D_2 =$ **0.5**

D4 Condition on D_2: **0.3**

D5 Plot: **0.6**

D6 $\Delta\omega =$ **0.2**

D7 $\tau_s =$ **0.2**